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SUPERSEDING
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15 December 1952

FEDERAL SPECIFICATION

INSECTICIDE, LIME-SULFUR, LIQUID CONCENTRATE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—Insecticide, lime-sulfur, liquid concentrate covered by this specification is suitable for the control of insects and fungi on trees, shrubs, and other growing plants.

1.2 **Classification.**—The insecticide shall be of but one type and one grade, as hereinafter specified.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

TT–P–141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Copies of Federal Specifications and the Index of Federal Specifications and Standards may be obtained upon application accompanied by check, money order, cash or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets SW., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Prices may be obtained from the Index of Federal Specifications and Standards or from the GSA Regional Offices. Single copies of this specification and other product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.)

2.2 Other publications.—The following publications, of the issues in effect on date of invitation for bids, form a part of this specification:

Governmental:

Federal Insecticide, Fungicide, and Rodenticide Act.

Nongovernmental:

Official Methods of Analysis of the Association of Official Agricultural Chemists.

(Copies of the Governmental publication referenced above may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.

(Official Methods of Analysis of the Association of Official Agricultural Chemists is published by the Association of Official Agricultural Chemists, P. O. Box 540, Benjamin Franklin Station, Washington 4, D. C.)

2.3 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 **Composition.**—Insecticide, lime-sulfur, liquid concentrate shall consist of an orange-red solution prepared by boiling lime, sulfur, and water together and shall have the following properties:

	Percent by weight		Test method paragraph
	Minimum	*Maximum*	
Calcium polysulfides	29	32	4.3.1
Calcium thiosulfate		2.5	4.3.2
Specific gravity at 60°/60°F.	1.270	1.295	4.3.3

3.2 **Workmanship.**—The finished product shall be clean, free from undissolved material and free from defects which may impair its utility.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—A sample of the insecticide sufficient to determine conformance of the material with this specification shall be taken at random from each lot of material. A lot shall consist of all material produced from the same

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materials and offered for delivery or delivered at the same time.

4.2 **Inspection.**—Inspection may be made at the site of manufacture, at any suitable point in transit, or after delivery to the point of destination, as specified in the contract or order. Inspection shall be made by a Government inspector to determine compliance with the requirements of this specification. Inspection shall also include packaging, packing, and marking for shipment.

4.3 **Tests.**

4.3.1 *Calcium polysulfides.*—The calcium polysulfide content shall be determined according to the methods for analysis of lime-sulfur solutions, Official Methods of Analysis of the Association of Official Agricultural Chemists.

Calculate calcium polysulfide as follows:

Calcium polysulfide = total calcium − (calcium as calcium thiosulfate + calcium as calcium sulfate) + total sulfur − (thiosulfate sulfur + sulfate sulfur).

4.3.2 *Calcium thiosulfate.*—The calcium thiosulfate content shall be determined according to the method for thiosulfate sulfur, Official Methods of Analysis of the Association of Official Agricultural Chemists. Calculate calcium thiosulfate from the amount of thiosulfate sulfur found.

4.3.3 *Specific gravity.*—The specific gravity of the insecticide shall be determined by method 418.3 of Federal Specification TT–P–141.

5. PREPARATION FOR DELIVERY

5.1 **Packaging and packing.**—Unless otherwise specified, the subject commodity shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate to the point of delivery.

5.2 **Marking.**

5.2.1 *Labeling.*—The label data shall be lithographed or stenciled on container body with a weather-resistant and non-corrosive black ink. Where plain olive-drab drums are employed a combination consisting of a stenciled label and a paper label may be used. The paper label shall be printed on a label made of strong paper affixed to the drum body with a water insoluble, non-corrosive adhesive and coated overall with a weather-resistant lacquer or enamel. The name of the product shall be stenciled on container body when paper labels are used. The product shall be registered and labeled in compliance with the Federal Insecticide, Fungicide and Rodenticide Act. The label data shall read as follows:

INSECTICIDE, LIME-SULFUR, LIQUID CONCENTRATE

Stock No. ----------

Active ingrediants	*Percent by weight*
Calcium polysulfide	—
Calcium thiosulfate	—
Inert ingredients	—
Total	100

Manufacturer's name and address
Manufacturer's batch No.
Net contents
Date of pack (month and year)

INSTRUCTIONS FOR USE

On trees, shrubs, and other plants for which lime-sulphur solution is used.

Use to control scale insects and spider mites, like the red spider.

Dormant spray.

It is most effective as a scalecide if applied during the dormant or winter season and when the temperatures are above freezing, preferably in the early spring. At such time dilute 10 gallons of lime-sulfur concentrate with 90 gallons of water.

For small quantities dilute 1 pint with water to make 1 gallon of spray.

Summer spray and fungicide.

To control red spiders and related mites and fungus diseases on plant foliage, dilute 2 gallons of lime-sulfur concentrate with 98 gallons of water. For small quantities use 2½ ounces (5 tablespoonfuls) to 1 gallon of water.

Caution: To be used only by trained personnel. Lime-sulfur will stain paints, brick or cement work unless it is protected by canvas or other covering while the application is being made. Do not use oil sprays on plants that have recently been sprayed with sulfur-con-

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taining insecticides. Use lime-sulfur solution with care during hot weather to avoid plant injury. To avoid getting sulfur into the eyes and nose, wear goggles and a respirator.

6. NOTES

6.1 Ordering data. — Purchasers should specify the title, number, and date of this specification, the size and weight of sample unit, if required, and exercise any desired options offered herein. See 4.2, and 5.1.)

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 Transportation description.

6.3.1 *Transportation description applicable to this item is:*

Lime and sulphur solution.
Carload weight 40,000 pounds.
Truckload weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standards of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army and Air Force

7.2.1 *Applicable specifications and standards:*

Military Specifications:

MIL–C–124—Containers (Cans, Pails, and Drums) Metal (For Other Than Subsistence Items).

JAN–C–490—Cleaning and Preparation of Ferrous Metal Surfaces for Organic Protective Coatings.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking of Shipments.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Sampling and inspection.*—Sampling and inspection of the end item covered by this specification, and those components not covered by applicable specifications shall be in accordance with 7.2.2.1 and 7.2.2.2. Sampling and inspection of components entering into the manufacture of the item and covered by applicable specifications shall be in accordance with the specifications for such components. The Government reserves the right to perform sampling and inspection for any requirements of this specification which are not specifically included in 7.2.2.1, and 7.2.2.2.

7.2.2.1 *Sampling.*

7.2.2.1.1 *Sampling for visual inspection of filled cans.*—Sampling for visual inspection shall be in accordance with Military Standard MIL–STD–105.

7.2.2.1.2 *Sampling for testing.*—The number of sample units to be used for testing of the end product shall be determined by the Government. Each sample unit shall contain ½ pint of the end item.

7.2.2.2 *Visual inspection.*

7.2.2.2.1 *Inspection of end product.*—Inspection of the end product shall be in accordance with the classification of defects and acceptable quality levels set forth below. The Government reserves the right to classify as a defect any departure from this specification not specifically listed below.

Defects of filled containers

Examine	*Defects*
Workmanship	Any seepage.
	Any breaks or dents affecting serviceability.

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Defects of filled containers—Continued

Examine	*Defects*
Type container	Not as specified.
Label data markings	Not clear; not legible; smears.
	Not lithographed with one print black ink on the outside of container body.
	Does not contain required markings as noted in 5.2.1.
Acceptable quality level (AQL).	2.5 defects per 100 cans.

7.2.2.2.2 *Inspection of net contents.*—Determination shall be made of the average net contents per can. The average shall not be less than specified.

7.2.3 *Packaging and packing (for domestic and overseas shipment).*—Unless otherwise specified not less than 5 gallons of the product shall be packaged in a round, closed top, liquid tight container constructed in accordance with the requirements of Military Specification MIL–C–124, type V. Container shall be fabricated throughout from not less than 24 gage phosphate-treated blackplate conforming to the requirements of grade I of Military Specification JAN–C–490. When phosphate-treated blackplate is not obtainable, the use of 24 gage cold rolled steel plate coated on one side with an interior coating, using interchemical interior coating 63A or equal, and meeting the requirements of Military Specification MIL–C–124, will be permitted.

7.2.3.1 Each container shall be fitted with a two inch threaded flange and plug type closure made as specified in Military Specification MIL–C–124.

7.2.3.2 Alternately, each container may be fitted with a 45 or 55 millimeter flange nozzle and leakproof snap-on cap.

7.2.3.3 Closures, overseals, and pouring spouts shall be made in accordance with Military Specification MIL–C–124, paragraph covering "snap-on (type which is opened by downward pressure in the center) closure." Closure shall have a commercial retractable type pouring spout suitable for the product. The stock number shall be embossed on the overseal.

7.2.4 Marking.

7.2.4.1 *Labeling.*—The label data specified in 5.2.1 shall be durably and legibly marked on the main panel of the body of the can by lithographing, stenciling or printing with a noncorrosive, weather-resistant, waterproof ink which will not smear nor rub off during handling.

7.2.4.2 *Shipping containers.*—Shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

U. S. GOVERNMENT PRINTING OFFICE: 1953

Copies of this specification may be purchased for 5 cents.

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